

PHY301-Circuit Theory lecture wise Question& Answers Prepared By
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Lecture no23:

Explain KCL equations?

KCL equation is written in nodal analysis to write nodal equation. To write KCL, we make use of Ohm's law in the form of $I=V/R$

At node (1)

$$V_1/3k + 4mA + 2mA + (V_1 - V_2)/6k = 0$$

To simplify this equation, we take L.C.M. 6 and multiply each term by 6 on both sides. So

$$6xV_1/3k + 6x 4mA + 6x 2mA + 6x (V_1 - V_2)/6k = 6x 0$$

Solving this we get

$$2V_1 + 24 + 12 + V_1 - V_2 = 0$$

$$3V_1 - V_2 = -36$$

At node (2)

$$V_2/4k + V_2/12k - 2mA + (V_2 - V_1)/6k = 0$$

To simplify this equation, we take L.C.M. 12 and multiply each term by 12 on both sides. So

$$12xV_2/4k + 12x V_2/12k - 12x 2mA + 12x (V_2 - V_1)/6k = 0$$

Solving this we get

$$3V_2 + V_2 - 24 + 2V_2 - 2V_1 = 0$$

$$6V_2 - 2V_1 = 24$$

Q)isMATRICES AND DETERMINANTS to solve the KCL and KVL equations is an easy and short method?

Matrices and determinant method is another technique used to solve simultaneously equations having different variables. However, you can use any convenient and simple method to solve such equations. It's on your choice.

Lecture no 24;

how to solve 3 matrix?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant is:

$$|A| = ad - bc$$

"The determinant of A equals a times d minus b times c"

It is easy to remember when you think of a cross:



- Blue means positive (+ad),
- Red means negative (-bc)

Example:

$$B = \begin{bmatrix} 4 & 6 \\ 3 & 8 \end{bmatrix}$$

$$\begin{aligned} |B| &= 4 \times 8 - 6 \times 3 \\ &= 32 - 18 \\ &= 14 \end{aligned}$$

For a 3×3 matrix (3 rows and 3 columns):

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

The determinant is:

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

"The determinant of A equals ... etc"

It may look complicated, but **there is a pattern**:

$$\left[\begin{array}{c|cc} a & e & f \\ \hline & h & i \end{array} \right] \times - \left[\begin{array}{c|cc} b & d & f \\ \hline & g & i \end{array} \right] + \left[\begin{array}{c|cc} c & d & e \\ \hline & g & h \end{array} \right] \times$$

To work out the determinant of a 3×3 matrix:

- Multiply **a** by the **determinant of the 2×2 matrix** that is **not** in **a**'s row or column.
- Likewise for **b**, and for **c**
- Add them up, but remember that **b** has a negative sign!

As a formula (remember the vertical bars $||$ mean "determinant of"):

As a formula (remember the vertical bars || mean "determinant of"):

$$|A| = a \cdot \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \cdot \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \cdot \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

"The determinant of A equals a times the determinant of ... etc"

Example:

$$C = \begin{bmatrix} 6 & 1 & 1 \\ 4 & -2 & 5 \\ 2 & 8 & 7 \end{bmatrix}$$

$$\begin{aligned} |C| &= 6 \times (-2 \times 7 - 5 \times 8) - 1 \times (4 \times 7 - 5 \times 2) + 1 \times (4 \times 8 - -2 \times 2) \\ &= 6 \times (-54) - 1 \times (18) + 1 \times (36) \\ &= -306 \end{aligned}$$

Lecture no 25:

Linear circuit

Linear circuits are proportional circuits having the output directly related to the input. In a linear circuit if we plot the Input vs Output of a circuit, it will be a straight line.

For example if we increase the value of applied voltage across a linear circuit, the current flowing in the circuit will increase proportional to the applied voltage.

Superposition Theorem

Dear Student,

The Superposition Theorem is an important concept in circuit analysis. It allows you to determine a voltage across a component or a branch current by calculating the effect of each source individually, and then algebraically adding each contribution.

Superposition may be considered for circuit analysis when:

There are two or more energy sources.

The sources are either voltage or current sources.

The circuit is not too complex.

There are five steps used in applying the Superposition theorem to a circuit.

1. Select one energy source.
2. Remove all other sources by
 - Replacing voltage sources with a short while retaining any internal source resistance.
 - Replacing current sources with an open while retaining any internal resistances.
3. Calculate the desired voltage drops or branch currents paying attention to the voltage polarities and current directions.
4. Repeat steps 1 through 3 for each other source individually.
5. Algebraically add the contributions of each voltage or current.

Source transformation

In a circuit containing dependent and independent sources, transformation is usually done by independent current or voltage sources keeping the dependent source constant.

Lecture no 26:

What is simulation package?

Simulation software is computer model based on the process of modeling a real or proposed system with a set of mathematical formulas. It is, essentially, a program that allows the user to do an operation through simulation without actually performing that operation.

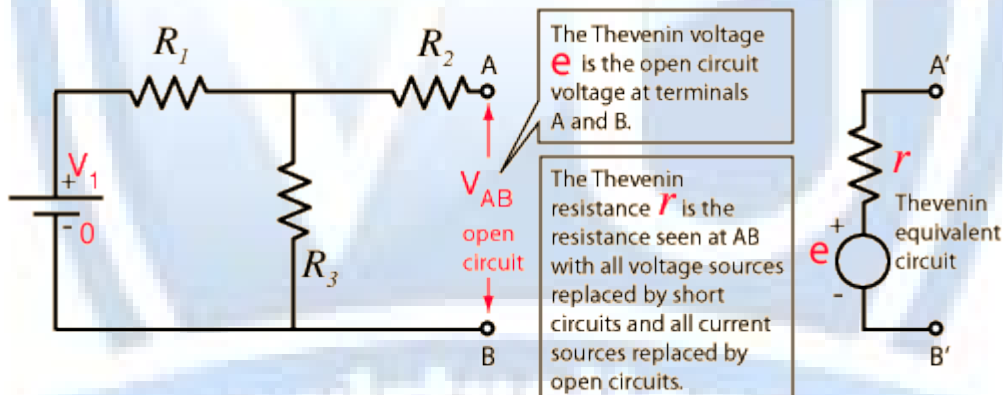
The simulation is time based, and takes into account all the resources and constraints involved, as well as the way these things interact with each other as time passes. Simulation also builds in the randomness you would see in real life. Simulation software is used widely to design equipment so that the final product will be as close to design specs as possible without expensive in process modification. Simulation software with real-time response is often used in gaming, but it also has important industrial applications. When the penalty for improper operation is costly, such as airplane pilots, nuclear power plant operators, or chemical plant operators, a mock up of the actual control panel is connected to a real-time simulation of the physical response, giving valuable training experience without fear of a disastrous outcome.

Applications of Thevenin Theorem?

in simplifying complex circuits and for different loads this theorem proven very useful

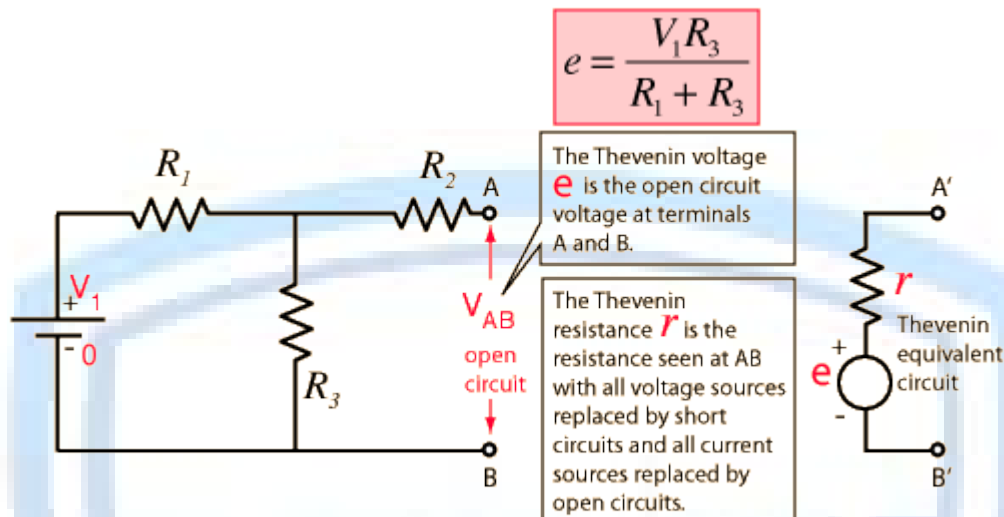
Thevenin's Theorem

Any combination of batteries and resistances with two terminals can be replaced by a single voltage source e and a single series resistor r . The value of e is the open circuit voltage at the terminals, and the value of r is e divided by the current with the terminals short circuited.



Thevenin Voltage

The Thevenin voltage e used in Thevenin's Theorem is an ideal voltage source equal to the open circuit voltage at the terminals. In the example below, the resistance R_2 does not affect this voltage and the resistances R_1 and R_3 form a voltage divider, giving



Question:What is Thevenin's theorem ?**Answer:**

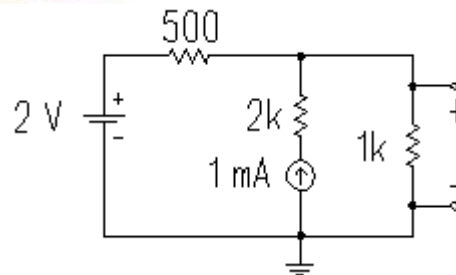
Thevenin's Theorem

The Thevenin's theorem states that any linear network can be represented by a voltage source in series with an impedance. We will generally use a simpler version: any circuit made up of voltage sources, current sources and resistors can be replaced by a voltage and a resistor.

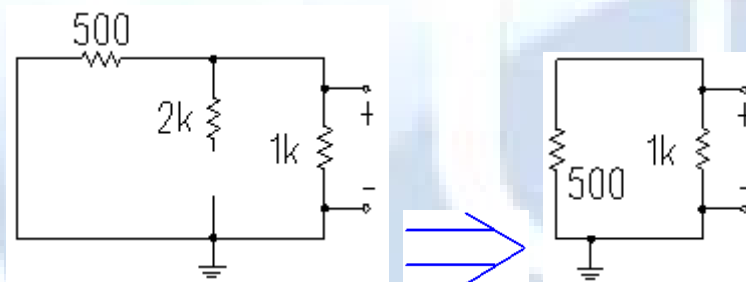
When applying the Thevenin Theorem there are three cases.

- **Case 1: Only independent sources** . In the typical case, there are no dependent sources in the circuit to be Thevenized. To find the Thevenin equivalent, first find the open circuit voltage, V_{oc} , this is the Thevenin voltage. To find the Thevenin resistance, set all sources to zero and find the resistance of the resulting circuit

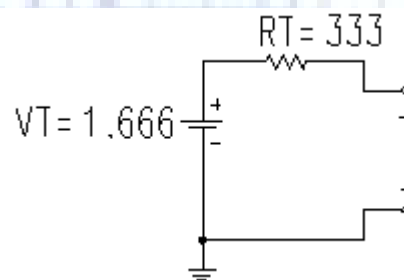
Consider again the circuit from above,



and try to find the Thevenin circuit at the terminals (i.e., across the 1k resistor). From the discussion of superposition, we know the open circuit voltage, V_{oc} , is 1.666 volts. The Thevenin resistance, R_T , is found by finding the equivalent resistance of the circuit with all source set to zero, as shown below



Obviously the Thevenin resistance, R_T , is $1k || 500 = 333\Omega$. Therefore the resulting circuit is:



- **Case 2: Independent and Dependent Sources.** If the circuit to be Thevenized has both dependent and independent source, the method described above cannot be used to find the Thevenin resistance. Instead, you must find the short circuit current,

I_{sc} (current through short circuit at terminals). Then the Thevenin resistance is given by $R_T = V_{oc} / I_{sc}$.

Lecture no 27:

Source Transformation Method?

source transformation is a method of converting voltage source into a current source or current source into a voltage source using the ohm's law.

Voltage source to current source conversion:

a voltage source connected in series with a resistance across 2 terminals gives $I = V/R$. this value of I is current source value which will be connected across the same 2 terminals in parallel to the same resistance

Current source to voltage source conversion:

a current source connected in parallel to a resistance across 2 terminals will give $V = IR$, this V is the value of Voltage source that will be connected across the same 2 terminals in series with the same resistance..

Conditions:

For source transformation it is required that voltage source is in series with a resistance or a current source is connected in parallel to the resistance.

Superposition method?

Superposition

Superposition is another algorithmic circuit analysis method, with a prescribed procedure, like the node-voltage or loop-current methods.

It applies only in circuits with 2 or more independent sources.

It comes from the mathematical notion of superposition in which a final solution can be built as the sum of two or more partial solutions. It works

well when the partial solutions can be obtained very easily. (Using voltage dividers or resistor reduction techniques.)

Superposition only applies to linear circuits, which are made up of linear circuit elements: resistors, voltage and current sources, capacitors, inductors, and linear amplifiers. It does not apply when the circuit has non-linear elements like diodes and transistors (electronic

What are the main concepts of Superposition Theorem?

The superposition theorem is a method of solving circuits, often used in circuits with more than one emf source. In a network containing multiple independent source, each source can be applied independently with the remaining source turned off. In order to use one source at a time, all other sources are "killed" temporarily. This means disabling the source so that it can not generate voltage or current, without changing the resistance of the circuit.

Keep these point in mind .

- (i) A voltage source such as a battery is killed or turned off by assuming a short circuit across its potential difference.
- (ii) A current source is killed or turned off by replacing it with an open circuit.
- (iii) The result obtained by applying each source independently are then added together algebraically to obtain a solution.

Lecture no 28:

Q)Difference between dependent and independent Current source or Volt source. ?

Dependent and Independent Sources:

There are two principal types of source, namely voltage source and current source. Sources can be either independent or dependent upon some other quantities.

An independent voltage source maintains a voltage (fixed or varying with time) which is not affected by any other quantity. Similarly an independent current source maintains a current (fixed or time-varying) which is unaffected by any other quantity.

Q) Difference between Norton Theorem and Thevenin's theorem basically?

THEVENIN'S THEOREM:

In solving any problem by Thevenin's theorem. We will follow the steps as stated below

First of all we will remove the load resistor (The resistor we want to calculate the unknown quantity that is voltage, current or power) from the circuit leaving behind an open circuit.

(2) Calculating the V_{th} at open terminals of the open circuit by any method.

(3) Calculating the R_{th} (The total resistance in the Thevenin's circuit) by open circuiting all current sources and short circuiting all voltage sources.

(4) After calculating V_{th} and R_{th} , re-inserting the load resistance R_L in the circuit in series with R_{th} and considering the V_{th} as a battery in series with these two resistances.

Q)NORTON'S THEOREM?

Working with Norton's theorem is same as working with Thevenin's theorem .

We will follow the same four steps.

But in second step instead of finding V_{th} we will find I_{Nor} by short circuiting the open terminals of the circuit.

In fourth step our circuit will consist of a current source of value equal to I_{Nor} and the

Norton resistance will become parallel with R_L .

Q)What is Norton's Theorem ?

- Norton's Theorem is a way to reduce a network to an equivalent circuit composed of a single current source, parallel resistance, and parallel load.
- Steps to follow for Norton's Theorem:
 - (1) Find the Norton source current by removing the load resistor from the original circuit and calculating current through a short (wire) jumping across the open connection points where the load resistor used to be.
 - (2) Find the Norton resistance by removing all power sources in the original circuit (voltage sources shorted and current sources open) and calculating total resistance between the open connection points.
 - (3) Draw the Norton equivalent circuit, with the Norton current source in parallel with the Norton resistance. The load resistor re-attaches between the two open

points of the equivalent circuit.

- (4) Analyze voltage and current for the load resistor following the rules for parallel circuits.

Q)What is peak voltage?

Peak voltage tell you how far the voltage swings, either positive or negative, from the point of reference. Peak voltage is only a moderately useful way of measuring voltage when trying to express the amount of work that will be done when driving a specified load. Some manufacturers use peak voltage to get the power output ratings of their amplifiers.

Lecture no 29:

Q)What is RMS voltage ?

RMS voltage is absolutely the most common way to measure/quantify AC voltage. It is also the most useful. Because AC voltage is constantly changing and is at or near the highest and lowest points in the cycle for only a tiny fraction of the cycle, the peak voltage is not a good way to determine how much work can be done by an AC power source (e.g. your amplifier, a wall outlet in your house...). DC voltage is constant. Its voltage level can be plugged directly into the formulas for power and you will get an accurate image of its ability to do work. RMS voltage will give you the same ability to predict how much work will be done by an AC voltage. The RMS voltage of a Pure sine wave is approximately $.707 \times \text{peak voltage}$. If you read voltage with a voltmeter you are generally given the RMS voltage of the wave form. Some meters display an 'average' voltage which is very close to RMS. When reading voltage with a voltmeter, the display indicates the RMS or average voltage not the peak or peak-peak voltage.

*If the waveform isn't a pure sine wave (like a square wave or a signal with mixed sine waves of different frequencies or music), multiplying the peak times $.707$ will not give an accurate RMS value and therefore will not give an accurate indication of the work that the waveform can produce when driving a load. For more complex signals, you need a meter that will calculate the RMS value from a set of samples taken at regular intervals.

Q)What is an Ideal diode?

The diode can be considered to be a one way street, that is it conducts electricity well in one direction but hardly any in the opposite direction. An ideal diode has no resistance in the forward direction and infinite resistance in the reverse direction. An ideal diode is like a light switch in your home. When the switch is closed, the circuit is completed; and the light turns on. When the switch is open, there is no current and the light is off.

However, the diode has an additional property; it is unidirectional, i.e. current flows in only one direction (anode to cathode internally). When a forward voltage is applied, the diode conducts; and when a reverse voltage is applied, there is no conduction. A mechanical analogy is a rat trap, which allows motion in one direction only.

Lecture no 30:

What is working principle of PN Junction Diode?

Assuming that you are familiar with p-type and n-type doped semiconductor materials, we will consider silicon, in simple terms.

lattice structure of the p-type, makes it a willing acceptor of the electrons from the n-type at the other side of the barrier.

This transfer through the junction requires a specific force, in electrical terms what we call voltage. For silicon 0.6 to 0.7 volts gets full and safe current flow through the junction.

Less than about 0.6 V the current will be much less.

If the voltage polarity bias of the

junction, is such that the pressure is now trying to force electrons from p to n, there will be only a tiny leakage current. The result can be seen to be that a diode is a one way valve for electrons. They can go from n to p but not p to n.

What is source transformation?

Source Transformation method is used to find unknown value of a voltage or current in a circuit.

If we have any source embedded within a network, say this source is a current source having a value I & there exists a resistance having a value R , in parallel to it. We can replace it with a voltage source of value $V=IR$ in series with same resistance R .

The reverse is also true that is a voltage source V , in series with a resistance R can be replaced by a current source having a value $I=V/R$

In parallel to the resistance R .

Parameters within circuit, for example an output voltage remain unchanged under these transformations.

Lecture no 31:

What is a function of diode?

Diodes allow electricity to flow in only one direction. The arrow of the circuit symbol shows the direction in which the current can flow. Diodes are the electrical version of a valve and early diodes were actually called valves. To see the figure please click [here](#). A diode is made of two different types of semiconductors right next to each other. One side is easy for electrons to travel through; one side is much tougher. It's something like trying to swim through a pool filled with

water and then a pool filled with mud: swimming through water is easy; swimming through mud is next to impossible. To an electron some semiconductors seem like water, some like mud. One side of the semiconductor boundary is like mud, one like water. If you try to get electricity to move from the mud side to the water side, there's no problem. The electrons just jump across the boundary, forming a current. But try to make electricity go the other way and nothing will happen. Electrons that didn't have to work hard to travel around the water side just don't have enough energy to make it into the mud side. (In real life, there are always a few electrons that can trickle in the wrong direction, but not enough to make a big difference.

There are a number of different electronic devices which tend to be called diodes. Although they're made differently they all have three things in common.

1. They have two leads like a resistor.
2. The current they pass depends upon the voltage between the leads.
3. They do not obey Ohm's law!

As an example we will use a typical diode called a pn-junction. This allows us to explain behavior of diodes. Remember, however, that there are other sorts of diodes which are built differently but show the same general behavior.

We create a pn-junction by joining together two pieces of semiconductor, one doped n-type, the other p-type. This causes a depletion zone to form around the junction (the join) between the two materials. This zone controls the behavior of the diode.

What is meant by Forward Voltage Drop ?

Electricity uses up a little energy pushing its way through the diode, rather like a person pushing through a door with a spring. This means that there is a small voltage across a conducting diode, it is called the forward voltage drop and is about 0.7V for all normal diodes which are made from silicon.

Lecture no 32:

How to draw a graph of 45 degree?

we are required to draw relation between Supplied voltage V_1 and voltage drop V_o across resistance R in form of graph.

The straight line in graph (at 45 degree) shows that V_1 and V_o are proportional to each other, which means that by increasing value of the supplied voltage V_1 , currents in resistance R also increases, due to which voltage V_o across that resistance also increases.

Q)what is mobility carrier?

The charge carriers electrons and protons (holes) in semiconductor that take part in current flow are called mobility carriers.

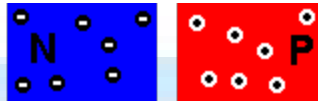
Lecture no 33:

Q)what is space region and diode current?

PN Junction and Space charge region.

We know that it is possible to turn a crystal of pure silicon into a moderately good electrical conductor by adding an impurity such as arsenic or phosphorus (for an N-type semiconductor) or aluminum or gallium (for a P-type semiconductor). By itself, however, a single type of semiconductor material isn't very useful. Useful applications start to happen only when a single semiconductor crystal contains both P-type and N-type regions. Here we will examine the properties of a single silicon crystal which is half N-type and half P-type.

Consider the silicon crystal represented in the figure below. Half is N-type while the other half is P-type. We've shown the two types separated slightly, as if they were two separate crystals.



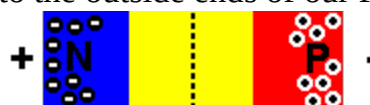
The free electrons in the N-type crystal are represented by small black circles with a "-" sign inside to indicate their polarity. The holes in the P-type crystal are shown as small white circles with a "+" inside. In the real world, it isn't possible to join two such crystals together usefully. Therefore, a practical PN junction can only be created by inserting different impurities into different parts of a single crystal. So let's see what happens when we join the N- and P-type crystals together, so that the result is one crystal with a sharp boundary between the two types.

You might think that, left to itself, it would just sit there. However, this is not the case. Instead, an interesting interaction occurs at the junction. The extra electrons in the N region will seek to lose energy by filling the holes in the P region.



This leaves an empty zone, or **Space region** (*depletion region*) as it is called, around the junction as shown to the right. This action also leaves a small electrical imbalance inside the crystal. The N region is missing some electrons so it has a positive charge. Those electrons have migrated to fill holes in the P region, which therefore has a negative charge. This electrical imbalance amounts to about 0.3 volt in a germanium crystal, and about 0.65 to 0.7 volt in a silicon crystal. This will vary somewhat depending on the concentration of the impurities on either side of the junction.

Unfortunately, it is not possible to exploit this electrical imbalance as a power source; it doesn't work that way. However, we can apply an external voltage to the crystal and see what happens in response. Let's take a look at the possibilities. Suppose we apply a voltage to the outside ends of our PN crystal.



We have two choices. In this case, the positive voltage is applied to the N-type material. In response, we see that the positive voltage applied to the N-type material attracts any free electrons towards the end of the crystal and away from the junction, while the negative voltage applied to the P-type end attracts holes away from the junction on this end. The result is that all available current carriers are attracted away from the junction, and the depletion region grows correspondingly larger. There is no current flow through the crystal because all available current carriers are attracted away from the junction, and cannot cross. (We are here considering an ideal crystal -- in real life, the crystal can't be perfect, and some leakage current does flow.) This is known as *reverse bias* applied to the semiconductor crystal.

Here the applied voltage polarities have been reversed.



Now, the negative voltage applied to the N-type end pushes electrons towards the junction, while the positive voltage at the P-type end pushes holes towards the junction. This has the effect of shrinking the depletion region. As the applied voltage exceeds the internal electrical imbalance, current carriers of both types can cross the junction into the opposite ends of the crystal. Now, electrons in the P-type end are attracted to the positive applied voltage, while holes in the N-type end are attracted to the negative applied voltage. This is the condition of *forward bias*. Because of this behavior, an electrical current can flow through the junction in the forward direction, but not in the reverse direction. This is the basic nature of an ordinary semiconductor diode.

It is important to realize that holes exist only within the crystal. A hole reaching the negative terminal of the crystal is filled by an electron from the power source and simply disappears. At the positive terminal, the power supply attracts an electron out of the crystal, leaving a hole behind to move through the crystal toward the junction again.

In some literature, you might see the N-type connection designated the *cathode* of the diode, while the P-type connection is called the *anode*. These designations come from the days of vacuum tubes, but are still in use. Electrons always move from cathode to anode inside the diode.

One point that needs to be recognized is that there is a limit to the magnitude of the reverse voltage that can be applied to any PN junction. As the applied reverse voltage increases, the depletion region continues to expand. If either end of the depletion region approaches its electrical contact too closely, the applied voltage has become high enough to generate an electrical arc straight through the crystal. This will destroy the diode.

It is also possible to allow too much current to flow through the diode in the forward direction. The crystal is not a perfect conductor, remember; it does exhibit some resistance. Heavy current flow will generate some heat within that resistance. If the resulting temperature gets too high, the semiconductor crystal will actually melt, again destroying its usefulness.

Always be careful to pay attention to the maximum specifications of a diode, and be sure to keep the operating conditions of the diode well within the indicated limits.

Lecture no 34:

Q)what is rectifier?

an electrical device which converts an alternating current into a direct one by allowing a current to flow through it in one direction only.

Q)Differences between small signal and large signal

A large signal is a DC signal (or an AC signal at a point in time) (one or more orders of magnitude larger than the small signal) used to analyse a circuit

containing non-linear components and calculate an operating point (bias) of these components.

A small signal is an AC signal superimposed on a circuit containing a large signal. In analysis of the small signal's contribution to the circuit, the non-linear components are modeled as linear components.

Q)What do small-signal model do ?

The small-signal model tells you the AC characteristics around a biased point. It is used when you need the AC characteristics (gain, impedance) of the transistor so that you can calculate the AC gain or impedance of the circuit that incorporates that transistor.

Lecture no 35:

Small-Signal mode ?

Small-signal modeling is a common analysis technique in electrical engineering which is used to approximate the behavior of nonlinear devices with linear equations. This linearization is formed about the DC bias point of the device (that is, the voltage/current levels present when no signal is applied), and can be accurate for small excursions about this point.

What is meant by "Reverse Voltage" in a diode?

Reverse Voltage When a reverse voltage is applied a perfect diode does not conduct, but all real diodes leak a very tiny current of a few μA or less. This can be ignored in most circuits because it will be very much smaller than the current flowing in the forward direction. However, all diodes have a maximum reverse voltage (usually 50V or more) and if this is exceeded the diode will fail and pass a large current in the reverse direction, this is called breakdown.

Lecture no36:

Q)What is diode?

A diode is simply a PN junction having two electrodes, Anode (+ve side) and a cathode (-ve side). No electrons move through the PN junction at equilibrium.

Generally the term bias refers to the use of a dc voltage to establish certain operating conditions for an electronic device. In relation to a diode, there are two bias conditions: forward and reverse biasing. Either of these bias conditions is established by connecting a sufficient dc voltage of the proper polarity across the PN junction.

There are two types of biasing generally used to operate the diode:

1. *Forward Biasing*
2. *Reverse Biasing*

Q)Diode current equation?

Diode current equation is

$$I = I_s e^{V_D / n V_T} \text{----- (A)}$$

V_D is diode voltage.

$$V_T = 25 \text{ mV} = 0.025 \text{ V}$$

For current I_1 and I_2 and voltage V_1 and voltage V_2

$$I_1 = I_s e^{V_1 / n V_T} \text{----- (B)}$$

$$I_2 = I_s e^{V_2 / n V_T} \text{----- (C)}$$

Taking natural log of equation (B) & (C) and after taking natural log subtract equation (C) from (B), we have:

$$V_2 - V_1 = n V_T \ln(I_2 / I_1)$$

Lecture 37:

Forward Biasing:

For forward biasing

The negative side of V_{Bias} is connected to the n region of the diode and the positive side to the p region. This is one requirement for forward bias. A second requirement is that the bias voltage, V_{Bias} , must be greater than the *barrier potential*.

To bias a diode, we apply dc voltage across it. Forward bias is the condition that allows current through the PN junction as shown in below figure, a dc voltage source connected by conductive material (contacts and wire) across a diode in the direction to produce forward bias. This external bias voltage is designated as V_{Bias} . The resistor, R , limits the current to a value that will not damage the diode.

Reverse Biasing:

Reverse Biasing is the condition that essentially prevents current through the diode, as we can see below in figure that a dc voltage source connected across a diode in the direction to produce reverse biasing. This external Bias voltage is designated as V_{BIAS} just as it was for forward bias. Notice that the positive side of V_{BIAS} is connected to the n region of the diode and the negative side is connected to the p region.

Lecture no 38:

What is passive sign Convention?

The passive sign convention: When we observe that positive current enters the positive terminal of a component, we say that the component obeys the passive sign convention (PSC). Therefore, when the passive sign convention is being obeyed, it indicates that a component is dissipating energy (or power) as charge is being displaced from a higher potential to a lower potential. One way to think about this is using another water analogy. If we splice a garden hose in a length of fire hose we create a pressure potential. The high pressure side is the one where the water is entering. The garden hose is analogous to a resistor which impedes the flow of current. Thus, when current flows through a resistor, a higher voltage potential will exist on the incoming

current side. In this example, the garden hose and the resistor are obeying the passive sign convention. To reiterate, the PSC is obeyed when the current enters an element's positive terminal and exits at the negative terminal. The passive sign convention is usually used for assigning reference marks for voltage drops across, and currents through, resistors, but we frequently assign a current and associated direction for a current through a voltage source in violation of the PSC. By the same token, we frequently define a voltage and its associated reference marks across a current source in violation of the PSC. To successfully apply Ohms law, you must consistently observe the proper relationship between applied voltage and the direction of current flow.

What is resistance?

Resistance

Resistance is a term that describes the forces that oppose the flow of electron current in a conductor. All materials naturally contain some resistance to the flow of electron current. We have not found a way to make conductors that do not have some resistance.

If we use our water analogy to help picture resistance, think of a hose that is partially plugged with sand. The sand will slow the flow of water in the hose. We can say that the plugged hose has more resistance to water flow than does an unplugged hose. If we want to get more water out of the hose, we would need to turn up the water pressure at the hydrant. The same is true with electricity. Materials with low resistance let electricity flow easily. Materials with higher resistance require more voltage (EMF) to make the electricity flow.

The scientific definition of one **ohm** is the amount of electrical resistance that exists in an electrical circuit when one amp of current is flowing with one **volt** being applied to the circuit.

Resistance depends upon the following factors

- a) length
- b) area
- c) temperature

$$R = \rho * L/A$$

Lecture no 39:

Is resistance good or bad?

Resistance can be both good and bad. If we are trying to transmit electricity from one place to another through a conductor, resistance is undesirable in the conductor. Resistance causes some of the electrical energy to turn into heat so some electrical energy is lost along the way.

However, it is resistance that allows us to use electricity for heat and light. The heat that is generated from electric heaters or the light that we get from light bulbs is due to resistance. In a light bulb, the electricity flowing through the filament, or the tiny wires inside the bulb, cause them to glow white hot. If all the oxygen were not removed from inside the bulb, the wires would burn up.

An important point to mention here is that the resistance is higher in smaller wires. Therefore, if the voltage or EMF is high, too much current will follow through small wires and make them hot. In some cases hot enough to cause a fire or even explode. Therefore, it is sometimes useful to add components called **resistors** into an electrical circuit to slow the flow of electricity and protect of the components in the circuit.

Resistance is also good because it gives us a way to shield ourselves from the harmful energy of electricity.

What is inductance?

INDUCTANCE:

The characteristic of an electrical circuit that opposes a change in current. The reaction (opposition) is caused by the creation or destruction of a magnetic field. When current starts to flow, magnetic lines of force are created. These lines of force cut the conductor inducing a counter emf in a direction that opposes current.

Lecture no 40:

What is a Step Up Transformer?

A step-up transformer is one whose secondary voltage is greater than its primary voltage. This kind of transformer "steps up" the voltage applied to it. For instance, a step up transformer is needed to use a 220v product in a country with a 110v supply.

What is a Step Down Transformer?

Its the opposite of the above, and would be used to run for example a 110v product in a country with a 220v mains supply

What 'Rating' Transformer do i need?

This depends entirely on the products you will be using it with. Give the electrical retailer a detailed list of all the products you will be using with it - and also their maximum outputs. From this information he/she will be able to advise the correct transformer rating needed.

A transformer converts alternating current (AC) from one voltage to another voltage. It has no moving parts and works on a magnetic induction principle; it can be designed to "step-up" or "step-down" voltage. So a step up transformer increases the voltage and a step down transformer decreases the voltage.

Q) What is Half Wave Rectifier?

A rectifier is a device that converts an alternating current, which periodically changes polarity, to direct current, which flows in only one direction. The half wave rectifier allows only one half of the alternating current to pass while the other half is blocked. This is achieved by using only a single diode.

Lecture no 41:

Q) What is INVERTING/Negative HALF-WAVE RECTIFIER?

During the negative half-cycle of a sinewave input, the output should be *positive*. During the positive half-cycle, the output should be *zero*. The circuit that accomplishes this amazing feat looks like the inverting amplifier with a couple of diodes added. Why two? Only one actually does the rectifying action. The other simply keeps the op amp in control while the signal output holds at zero.

Q)What is the Difference between half wave and full wave rectification?

1. Half-wave rectification



A half wave rectifier is a special case of a clipper. In half wave rectification, either the positive or negative half of the AC wave is passed easily, while the other half is blocked, depending on the polarity of the rectifier. Because only one half of the input waveform reaches the output, it is very inefficient if used for power transfer. Half-wave rectification can be achieved with a single diode in a one phase supply.

In half wave rectification, either the positive or negative half of the AC wave is passed, while the other half is blocked. Because only one half of the input waveform reaches the output, it is very inefficient if used for power transfer. Half-wave rectification can be achieved with a single diode in a one phase supply, or with three diodes in a three-phase supply.

in half wave rectifier only one diode is there. The diode will be forward biased in the positive half cycle and reverse biased in the negative half cycle. So the out put will be available only during the positive half cycle and the out put will be unidirectional.

Full-wave rectification

A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to DC (direct current), and is more efficient. However, in a circuit with a non-center tapped transformer, four diodes are required instead of the one needed for half-wave rectification. (See semiconductors, diode). Four rectifiers arranged this way are called a diode bridge or bridge rectifier:

For single-phase AC, if the transformer is center-tapped, then two diodes back-to-back (i.e. anodes-to-anode or cathode-to-cathode) form a full-wave rectifier (in this case, the voltage is half of that for the non-tapped bridge circuit above, and the diagram voltages are not to scale).

In a full wave rectifier during each half cycle one of the diode will be forward biased and output will be available across the load resistor. So we get continuous dc output throughout the cycle.

Lecture no 42:

PEAK INVERSE VOLTAGE:

□ When one of the diodes in a full wave rectifier is reverse biased, the voltage across it is approximately equal to V_2

.The peak load voltage supplied by the full wave rectifier is equal to the one half of the secondary voltage V_2

.Therefore, the reverse voltage will be twice the peak load voltage

Q) What is FULL WAVE BRIDGE RECTIFIER:?

the bridge rectifier is the most commonly used full wave rectifier circuit for several reasons

(1) It does not require the use of center-tapped transformer, and therefore can be coupled directly to the ac power line, if desired.

(2) Using a transformer with the same secondary voltage produces a peak output voltage that is nearly double the voltage of the full wave center-tapped rectifier. This results in the higher dc voltage from the supply.

Lecture no 43:

What is Power dissipation?

Power dissipation

When a current flows through a component, that component will heat up. This process is called power dissipation and is measured in Watts. The power dissipation of a device can be calculated very easily:

$$P=V \cdot I$$

Let's calculate the power dissipation of a 100 ohms resistor connected to a 9V battery. The voltage across the resistor will be 9V. The current is $9V/100\text{ohms} = 90\text{mA}$. So the power dissipation will be:

$$9V \cdot 90\text{mA} = 810\text{mW}$$

It is very important to calculate the power dissipation of the components in your design. A regular resistor has a maximum dissipation rating of 0.25W (= 250mW). If you would have used

such a resistor in the example above, it would have blown. A 1W resistor is a good choice. Since it's so important, let's create an equation with which we can easily calculate the power dissipation of a resistor. We know:

$$P = V \cdot I \text{ ----(1)}$$

$$V = I \cdot R \text{ ----(2)}$$

$$I = V/R \text{ ----(3)}$$

Substituting (2) in (1) and (3) in (1) respectively results in:

$$P = I^2 \cdot R$$

$$P = V^2 / R$$

With these equations you can easily calculate the power dissipation when you connect a DC voltage source to a resistor.

Lecture no 44:

OPERATING CONFIGURATIONS OF BJT:

□ There are three basic configurations under which transistor may be operated; these configurations depend upon the situation and requirements.

These configurations are called

1. Common Base Configuration.
2. Common Emitter Configuration.
3. Common Collector Configuration.

What is meant by " Branch " ?

Branches A branch is any path in the circuit that has a node at each end and contains at least one voltage source or resistor but contains no other nodes. This circuit contains 6 branches, denoted B1, ..., B6. If branch B4 did not contain a resistor then it could be deleted and nodes N2 and N3 could be considered one and the same node

Lecture no 45:

MODES OF OPERATION OF BJT:

- ☐ There are three modes of operation for BJT

1. Cut Off

When both the PN-junction of transistor are reverse –biased, the resulting mode of operation of the transistor is known as cut-off or simply as off-mode. This means that the BJT will be “OFF” and no current will flow through it. The fully supply voltage will exist across it. In everyday life an off switch will not permit any current to flow through an electric bulb and the full supply voltage will be present across the switch.

2. Active Mode

When BEJ of a transistor is forward-biased but BCJ is reverse biased the mode of operation of a transistor is referred to as its active-mode. A bipolar junction transistor in its active-mode of operation is called as an amplifier and is able to amplify voltage or current signals both.

3. Saturation

If both the PN-junctions, that is, the base-emitter junction (BEJ) and the base-collector junction (BCJ) are forward-biased, the transistor is said to be in its saturation (or ON) mode of operation.

MODES OF OPERATION OF BJT:

EBJ CBJ

- ☐ Cut Off Reverse Reverse
- ☐ Active Forward Reverse
- ☐ Saturation Forward Forward

For the transistors shown, state mode of operation.

Q)What are the rules for parallel and series resistances in a network or circuit?

Keep these two things in mind when you are solving such circuits.

Resistors are in series if they carry exactly the same current (share one common node).

Resistors are in parallel if they have the same voltage across them and are connected exactly between the same two nodes.

